CHAPTER 4

RESULTS AND DISCUSSION

Chapter 4 reports the results of the MedShield Decision Support System and examines how far the available evidence answers the capstone questions. The clearest result is a traceable historical sales layer connected to a dashboard. A candidate population of 37,728 accepted sales facts covers 2017 to 2025. Missing months remain visible, while product, area, and forecast displays identify their evidence limits. Predictive publication, external-signal forecasting, and operational inventory optimization remain incomplete; the chapter treats evaluated features separately from measured business outcomes.

The discussion follows the project objectives and the guiding question: How can MedShield use historical sales data, validated external signals, and scenario-based planning to improve demand visibility and reduce expiry-driven inventory losses for the 2026 planning cycle? Observed sales, candidate mappings, forecast evaluations, weather proxies, and planning scenarios carry different levels of evidence. The distinctions matter because a working screen does not establish an approved model, a verified stock position, or a reduction in expired inventory. Published studies inform the choice of methods, while MedShield receipts provide the evidence for MedShield results (Nguyen et al., 2021; World Health Organization, 2024).

The current MedShield sales record ends in December 2025. For elapsed 2026 months without uploaded client sales, the dashboard marks recency-weighted same-calendar values as gap estimates. The estimates support provisional planning while 2026 actual records are pending. They remain separate from observed sales, cannot serve as a 2026 validation holdout, and require reassessment after actual records arrive. Future 2026 months remain unavailable rather than appearing as actual sales.

4.1 Presentation of Results

Section 4.1 presents the implemented system and the evidence behind its results before discussing the patterns. The objective audit, architecture, and data-quality profile establish which claims can be defended. Historical sales refer to observed 2017 to 2025 records. A 2026 value without an uploaded actual remains a separately labeled estimate pending the client record.

4.1.1 Alignment With the Research Questions and Objectives

The proposal defined five objectives: describe sales and seasonality, prioritize products and areas, forecast demand with external signals, support inventory decisions, and present the results through five dashboard modules. The implementation moved furthest on the first, second, and fifth objectives. Forecasting has reached controlled evaluation but not release approval. The prescriptive layer can solve constrained scenarios, but the study does not contain the stock, expiry, lead-time, supplier, ordering-cost, or holding-cost data needed to measure an actual reduction in wastage.

Table 4.1 Objective achievement audit

Objective	Evidence expected	Observed result	Assessment
SO1	Sales performance, year-over-year patterns, and STL seasonality	Historical summaries and a 72-month STL decomposition are available. Ten earlier months remain unobserved, and the full geography mapping still needs owner approval.	Partially achieved
SO2	Product, territory, and account priorities using Pareto ABC and predictive scoring	Deterministic Pareto and ABC candidate views are implemented. Product labels are not approved SKUs, and no XGBoost classifier is published.	Partially achieved
SO3	Territory forecasts using sales, DII, RSI, Prophet, and XGBoost	Sales-only baselines and offline candidates were evaluated. External joins have zero forecast-eligible rows, and no model is approved or activated.	Not achieved for operational use
SO4	EOQ, ROP, and allocation decisions aimed at expiry wastage of at most 5 percent	The interface supports reviewed allocation scenarios. Real inventory and expiry outcomes are absent, so the 5 percent target was not tested.	Scenario capability only
SO5	Five dashboard modules with upload validation and decision support	The five principal modules exist. Browser-to-warehouse CSV ingestion remains unfinished; Signal 2 alerts and collaborative filtering are not verified outputs.	Partially achieved

Source: Capstone proposal objectives and the MedShield current-state, analytics, and QA records as of 27 September 2026.

A completed interface does not by itself prove a research outcome. The forecast screen can display a baseline without an approved model, and the planning solver can operate on assumed inputs without verified stock. Each following result therefore pairs an implemented function with the limit of its evidence.

4.1.2 Implemented System and Runtime Architecture

MedShield was implemented as an authenticated web application. The Next.js frontend displays analytical views and sends requests to an Express and TypeScript gateway. The gateway performs validation, authorization, and Databricks query orchestration. Databricks Unity Catalog stores the analytical source of truth through Bronze, Silver, Audit, and Gold candidate schemas. Supabase supplies identity and audit support; it does not provide the dashboard measures. Python services are used for offline preparation, model experiments, and validation, but they are not a browser fallback.

The runtime boundary remains strict. If Databricks or a required view is unavailable, the gateway returns an error and the dashboard shows an unavailable state. Local JSON, seeded values, and old spreadsheets do not replace the missing result. The declared runtime source makes displayed analytical values traceable.

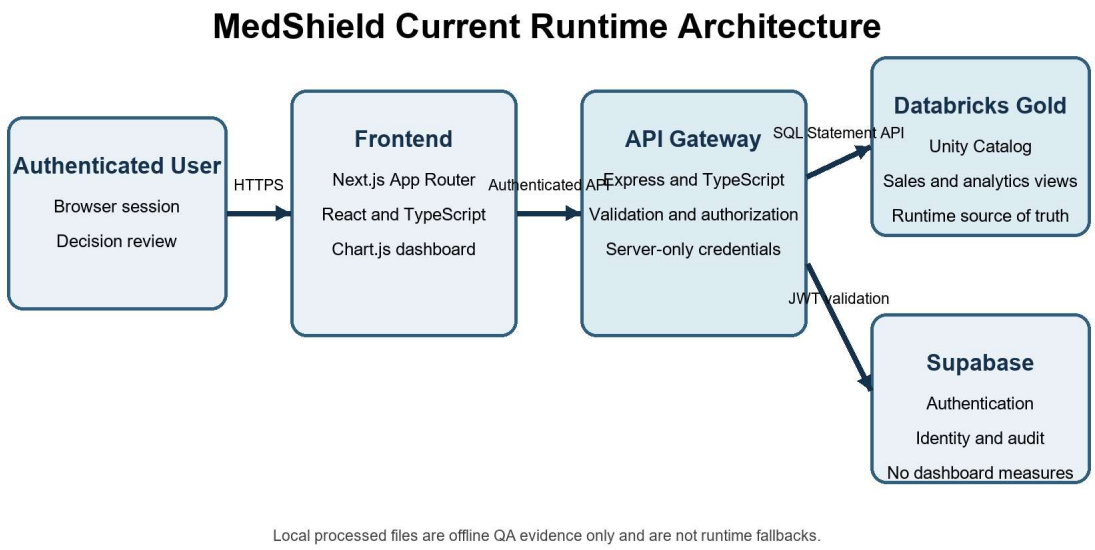


Figure 4.1 MedShield runtime architecture

Source: MedShield Architecture and Database documentation, September 2026.

Table 4.2 Implemented system layers

Layer	Implemented component	Responsibility	Boundary
Presentation	Next.js, React, TypeScript, Chart.js	Filters, charts, tables, exports, readiness labels, and scenario forms	Does not query Databricks or external providers directly
Gateway	Express and TypeScript	Authentication checks, parameter validation, query orchestration, and response contracts	Server-only credentials; explicit unavailable states
Analytical store	Databricks Unity Catalog	Bronze source evidence, Silver assessed records, Gold candidate facts and aggregates	Candidate publication pending business approval
Identity and audit	Supabase Auth and audit persistence	Session identity, roles, and application audit events	Not an analytical source
Offline analytics	Python jobs and tests	Reconciliation, STL, model evaluation, and scenario validation	Not a live dashboard dependency

Source: MedShield Architecture, Database, Security, and Project documentation.

4.1.3 Data Sources Coverage and Quality

The historical sales candidate covers 2017 to 2025 and contains 37,728 accepted fact rows. The proposal emphasized 2021 to 2025 and three Philippine regions, while the warehouse retains the longer history available in the supplied records. The broader period provides context. Regional interpretation remains conditional on approval of the area mapping. Department of Health and weather sources provide historical context but are not approved sales regressors.

Table 4.3 Data sources and current analytical status

Source	Coverage	Current use	Status and limitation

Source	Coverage	Current use	Status and limitation
MedShield sales	2017 to 2025 actuals; 2026 actuals pending	Historical demand, revenue, product, area, and baseline forecast evaluation	37,728 accepted facts; elapsed 2026 gaps may be estimated and remain separately labeled
DOH surveillance	2018 to 2026 declared; 2026 partial	Historical disease context and candidate regressors	Territory join not approved; no individual health inference
PAGASA observations	2017 to 2024	Official historical weather context	No official 2025 observations in the project; no approved sales match
NASA POWER	108 monthly rows for nine areas in 2025	Separately labeled weather proxy	Provider proxy, not PAGASA; 32 candidate matches do not approve regression
Operational inventory	Not supplied	Would support EOQ, ROP, safety stock, allocation, and expiry analysis	Scenario inputs only

Source: MedShield Current State, System Documentation, and publication-readiness evidence.

The quality review identified a complete 108-month calendar but only 98 months with observed transactions. The ten unavailable months are May, June, July, August, and October 2017; January and February 2018; and October, November, and December 2019. The gaps indicate missing observations, not zero demand. Null periods prevent trend and validation routines from creating artificial declines.

Table 4.4 Sales candidate quality profile

Quality measure	Observed value	Interpretation
Accepted fact rows	37,728	Checksum-bound candidate population
Revenue-eligible rows	36,848	Rows included in the net-sales measure
Margin-eligible rows	34,105	Rows with a valid margin denominator
Warning-bearing rows	32,749	Warnings remain visible and are not equivalent to rejection
Calendar coverage	108 months total; 98 observed; 10 missing	Missing periods remain unavailable rather than zero-filled
Product identity	4,561 source labels; 0 approved master entries	Product findings describe labels, not verified canonical SKUs
Area identity	22 derived labels	Some labels mix geography, customer type, or business line and require approval
Returns evidence	673 rows	Signed returns remain separate while contributing to net revenue

Source: Version-bound publication and QA evidence for the 37,728-row Databricks candidate.

4.2 Data Analysis

Section 4.2 interprets sales, seasonality, product and area priorities, forecast evaluations, and planning scenarios against the research objectives. Recorded performance remains separate

from model outputs and user assumptions. Estimates for elapsed 2026 months with pending actual records do not enter the observed 2017 to 2025 totals or serve as forecast-accuracy evidence.

In the QA review's default October 2025 to September 2026 window, observed sales were ₱61,082,570, estimated gaps were ₱88,804,257, and the combined display was ₱149,886,827, rounded to pesos. The combined amount is a mixed observed-and-estimated display value, not 2026 actual revenue or a forecast validation result. A later client upload can replace an estimate only after the sales pipeline and publication controls are completed (MedShield Capstone Group 9, 2026c).

4.2.1 Historical Sales Performance and Seasonality

Across the accepted candidate, observed net sales totaled ₱626,946,741.81. Workbook gross profit totaled ₱288,487,498.72. The weighted gross-margin rate was 60.26 percent, calculated only over the ₱478,720,670.03 margin-eligible revenue basis. The transaction-level workbook measure is not company net income. Dividing gross profit by all reported net sales would use the wrong denominator for the stated rate.

Table 4.5 Annual sales and workbook gross profit

Year	Transactions	Net sales	Workbook gross profit	Margin eligible sales
2017	1,978	₱10,211,167.35	₱6,316,157.68	₱9,212,606.10
2018	6,181	₱51,619,691.27	₱28,328,721.76	₱44,882,495.39
2019	5,446	₱62,542,304.49	₱29,125,469.95	₱41,211,281.42
2020	3,980	₱63,593,274.63	₱28,013,993.68	₱40,259,988.09
2021	3,407	₱77,860,206.26	₱50,420,770.93	₱77,804,247.26
2022	3,980	₱63,234,990.04	₱40,799,049.06	₱60,493,221.21
2023	5,381	₱62,228,966.85	₱33,585,582.48	₱59,822,843.64
2024	2,831	₱50,259,319.88	₱19,563,837.42	₱49,956,893.05
2025	4,544	₱185,396,821.04	₱52,333,915.76	₱95,077,093.87

Source: Local descriptive calculation receipt for Delta version 15, 26 September 2026. Current-state and QA summaries report matching overall totals for version 16; the version-16 calculation receipt was unavailable in the local checkout. Amounts are candidate values in Philippine pesos.

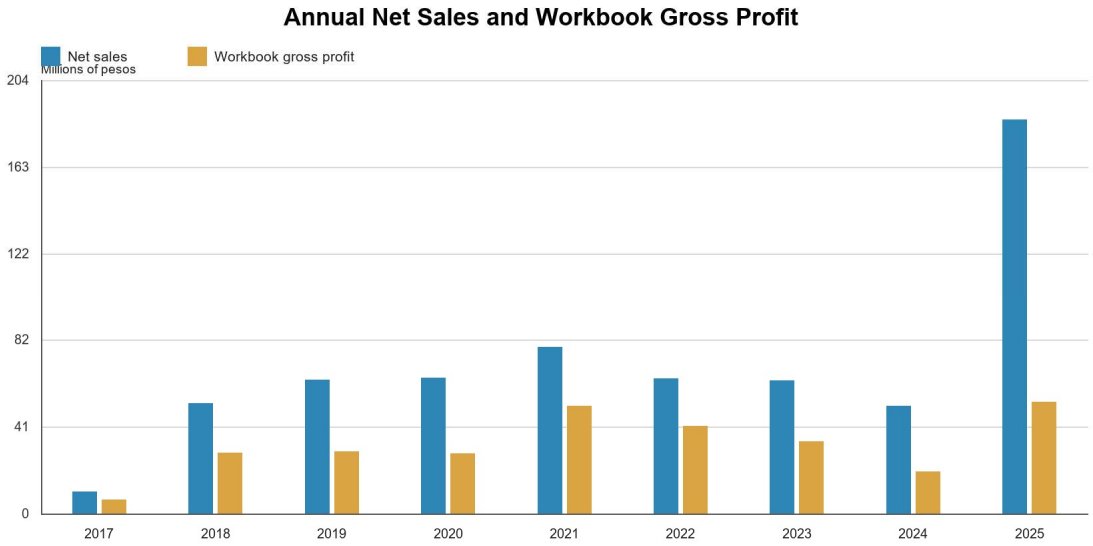


Figure 4.2 Annual net sales and workbook gross profit

Source: Generated from the version-bound descriptive calculation receipt.

The annual series records observed activity without proving equal completeness across years. The 2025 revenue increase is material in the candidate dataset, but source adjustments leave product attribution under review. The recorded financial total can be reconciled while product-level and causal explanations remain provisional.

From 2024 to 2025, recorded net sales increased from ₱50.26 million to ₱185.40 million, or about 268.9 percent, while workbook gross profit increased from ₱19.56 million to ₱52.33 million, or about 167.5 percent. Gross profit rose substantially, although its increase was smaller in proportion to revenue. The main measurable explanation is the available calculation basis: ₱49.96 million, or 99.4 percent, of 2024 net sales had fields required for the gross-margin calculation, compared with only ₱95.08 million, or 51.3 percent, of 2025 net sales. Within the matched basis, the calculated margin rate was approximately 39.2 percent in 2024 and 55.0 percent in 2025. The chart cannot establish higher costs, heavier discounts, or an unfavorable product mix as the cause. Much of 2025 revenue lacks the same margin-eligible basis, while source adjustments and product attribution remain under review. A causal explanation requires finance approval and complete matched cost evidence.

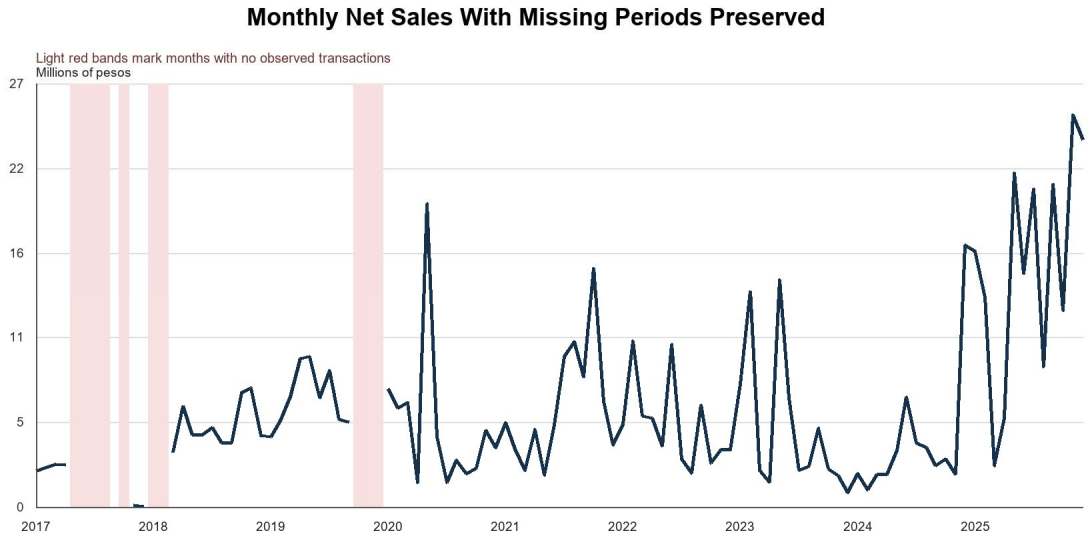


Figure 4.3 Monthly net sales and missing coverage

Source: Generated from the 108-month Databricks descriptive snapshot.

Observed calendar-month averages were highest in May at ₱9.87 million and June at ₱7.55 million. The pattern calls for seasonal review but does not show a disease or weather effect. Infectious diseases can follow seasonal cycles, yet a shared calendar pattern cannot establish a causal driver (Martinez, 2018). MedShield therefore presents month averages separately from its STL decomposition.

Table 4.6 Observed calendar month averages

Month	Average net sales	Observed years
Jan	₱6,405,176.08	8
Feb	₱7,098,050.31	8
Mar	₱3,879,490.79	9
Apr	₱4,433,381.20	9
May	₱9,871,031.23	8
Jun	₱7,553,552.93	8
Jul	₱6,832,765.69	8
Aug	₱5,074,209.17	8
Sep	₱6,073,372.18	9
Oct	₱6,538,044.40	7
Nov	₱6,502,031.32	8

Month	Average net sales	Observed years
Dec	₱7,126,212.47	8

Source: Descriptive snapshot. Method is observed calendar-month average, not STL.

The STL routine used a 12-month period, a seasonal window of 13, robust fitting, and no imputation. Early gaps and short runs left only January 2020 through December 2025 eligible. The 72-month decomposition reconstructs the observed series to numerical precision. STL separates trend, seasonal, and remaining variation for inspection (Cleveland et al., 1990); the MedShield plot does not establish a disease or weather cause.

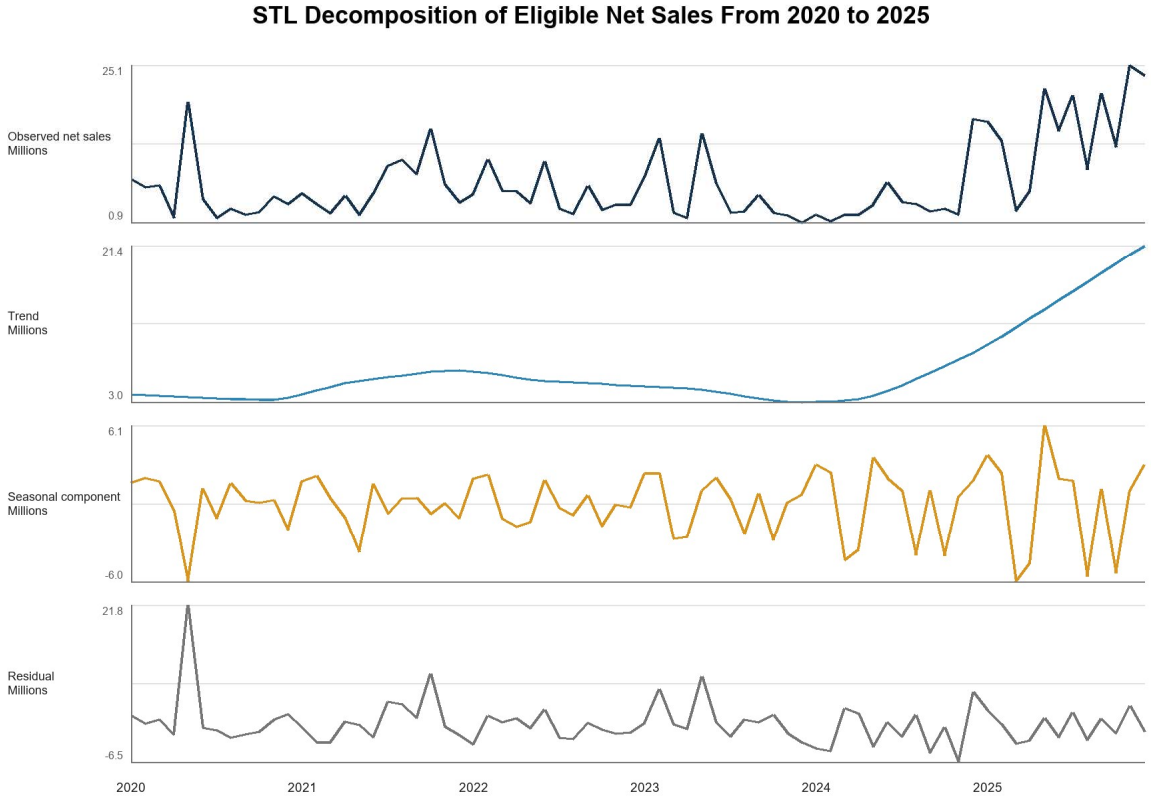


Figure 4.4 STL decomposition of eligible net sales

Source: Checksum-bound STL evidence generated with statsmodels 0.14.6.

4.2.2 Product and Area Prioritization

The second objective asks which products, territories, and accounts warrant attention. The descriptive layer answers through positive-sales contribution and deterministic Pareto or ABC grouping. No predictive XGBoost classifier has been approved. An ABC class describes contribution within selected history; it does not predict future urgency or stock-out risk.

Table 4.7 Top source product labels by net sales

Rank	Source product label	Net sales	Share	ABC
1	SPEEDA 2.5IU/0.5ML	₱22,466,448.23	3.57%	A
2	PROHAB 10MG/ML 20ML	₱16,530,532.11	2.63%	A
3	BUPIRIGHT 5MG/ML 0.5% IN 8% 4ML	₱15,966,455.39	2.54%	A
4	SITIXON 1G	₱13,051,271.68	2.08%	A
5	MONOWEL 1G IV	₱12,313,552.50	1.96%	A
6	PROFUREX 750MG	₱11,987,494.40	1.91%	A
7	EVAPROST 250MCG/ML	₱9,753,530.12	1.55%	A
8	F6 FETAL MONITOR SINGLE	₱7,848,998.50	1.25%	A
9	SPEEDA	₱7,558,970.97	1.20%	A
10	EUROXONE 1G	₱7,311,190.91	1.16%	A

Source: Delta version-15 descriptive receipt and Databricks product candidate logic. Share denominator is positive eligible product sales of approximately ₱628.57 million, not overall net sales of ₱626.95 million. Labels are not approved canonical SKUs.

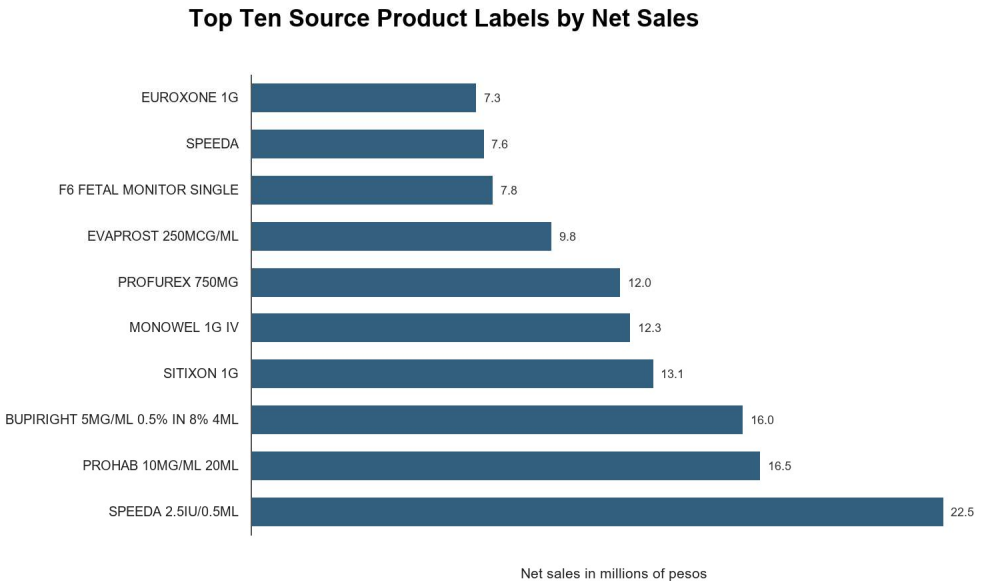


Figure 4.5 Top ten source product labels by net sales

Source: Generated from the descriptive calculation receipt.

SPEEDA 2.5IU/0.5ML ranked first with ₱22,466,448.23, or 3.57 percent of positive eligible product sales. The top five labels represented 12.78 percent and the top ten 19.85 percent on the same positive-sales basis. Overall net sales are lower because signed returns contribute to the net figure. The ranked labels show a dispersed portfolio and require a full-population Pareto calculation. Separate SPEEDA entries also suggest possible alias fragmentation. Multiple-criteria

ABC research offers a broader prioritization approach, but MedShield currently reports a deterministic revenue-based candidate rather than a validated multi-criteria classifier (Flores and Whybark, 1986).

Table 4.8 Top derived area labels by net sales

Rank	Derived label	Net sales	Share	Workbook gross profit
1	GOVERNMENT	₱310,629,101.40	49.55%	₱130,353,976.35
2	HOSPITAL	₱165,121,239.71	26.34%	₱69,103,315.46
3	BATANGAS	₱28,556,256.25	4.55%	₱17,838,380.41
4	QUEZON	₱27,207,048.54	4.34%	₱17,340,506.74
5	ADMIN	₱16,359,733.69	2.61%	₱8,706,600.10
6	EQUIPMENT	₱14,475,259.73	2.31%	₱7,679,164.32
7	LAGUNA	₱14,376,324.92	2.29%	₱8,509,063.56
8	MARINDUQUE	₱13,469,314.65	2.15%	₱6,984,765.63
9	CAM NORTE	₱11,638,151.07	1.86%	₱7,508,275.57
10	CAVITE	₱7,799,599.04	1.24%	₱5,155,855.62

Source: Databricks area candidate view. Mixed commercial labels; not a pure geography dimension.

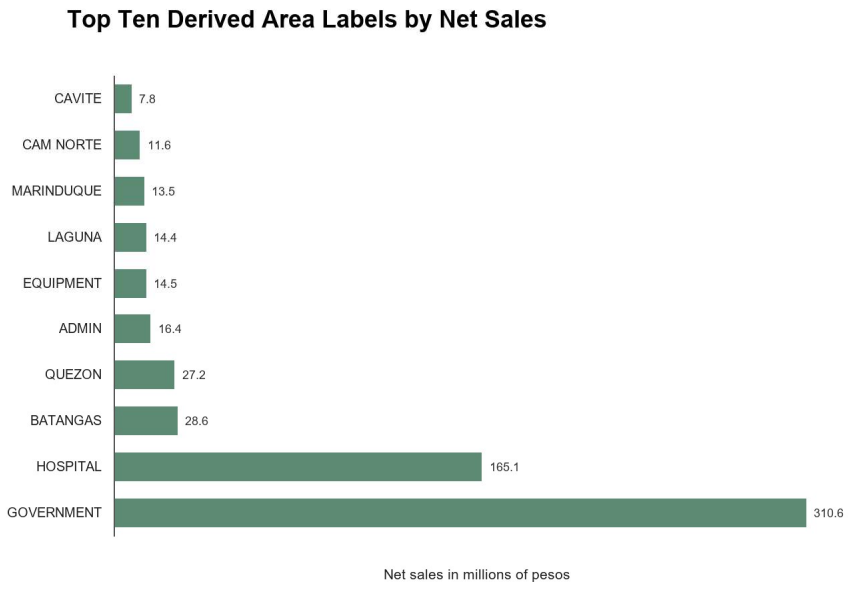


Figure 4.6 Top ten derived area labels by net sales

Source: Generated from the descriptive calculation receipt.

GOVERNMENT contributed 49.55 percent of candidate net sales and HOSPITAL contributed 26.34 percent. The labels describe commercial segments, not geographic territories. Batangas, Quezon, Laguna, Marinduque, Camarines Norte, and Cavite also appear in the legacy

area field. The dashboard exposes buyer-sector and geographic mappings separately, but final owner approval remains open. Government-labeled transactions are financially important in the current candidate; the label does not represent a verified territorial market share.

4.2.3 Forecast Evaluation and External Signals

The third objective was only partly realized. The system can evaluate simple sales-only baselines and persist candidate experiment evidence, but it has no approved operational forecast. In the fixed-origin Government net-sales comparison retained by the dashboard evidence, the last-observed-value method outperformed the seasonal-naive benchmark across the same nine observed holdout months. Both errors were high, and both methods underpredicted on average. The result supports transparent baseline comparison, not model promotion.

No 2026 MedShield sales actuals are present in the current Gold contract. The dashboard's 2026 gap estimates cannot serve as a holdout result or be compared with actual 2026 performance. The estimates maintain planning continuity while actual data are pending. Forecast validation remains retrospective and uses observed months within the 2017 to 2025 history.

Table 4.9 Fixed origin Government net sales benchmark

Method	Paired months	MAE	RMSE	WAPE	Bias
Seasonal naive	9	₱2,695,359.26	₱3,459,768.58	85.05%	-₱2,574,060.06
Last observed value	9	₱1,987,359.74	₱2,632,312.49	62.71%	-₱1,230,712.16

Source: Section 8 historical acceptance evidence. Missing actuals were excluded rather than set to zero.

A later offline experiment evaluated 15 configurations: four simple benchmarks, two ETS models, and nine Prophet configurations. Last value remained the development-selected candidate, with 83.20 percent development WAPE and 37.76 percent retrospective 2025 WAPE. No challenger met the promotion criteria, empirical uncertainty intervals were unsupported, and the model release pointer remained empty. Forecasting guidance requires accuracy checks on observations withheld from model fitting (Hyndman and Athanasopoulos, 2021). MedShield's retrospective errors therefore justify withholding an operational forecast even though a candidate method was selected.

Table 4.10 Predictive readiness audit

Evidence item	Observed result	Interpretation
Candidate configurations	15 evaluated	Experiment completed and persisted as candidate evidence

Evidence item	Observed result	Interpretation
Development selection	Last value; WAPE 83.20%	Selected among tested candidates but high error
Retrospective 2025 audit	Last value; WAPE 37.76%	Historical audit only; not a 2026 holdout
Intervals	Unsupported	Insufficient lead-specific error history for defensible bands
DOH and PAGASA forecast rows	0 forecast eligible	External regression remains blocked
Publication	No active model release	No operational forecast may be claimed

Source: MedShield Analytics, Predictive Operations, Current State, and QA records dated 27 September 2026.

The external data audit reinforces the forecast boundary. The current candidate contains 2,171 DOH and PAGASA rows, but none are join-ready or forecast-eligible. NASA POWER supplies separately labeled proxy observations and 32 candidate sales matches; the matches do not establish an approved geographic join or predictive benefit. No evidence supports a claim of improved MedShield forecasts from rainfall, disease intensity, or typhoon signals. Current source-identity and publication gates supersede older exploratory validation labels.

4.2.4 Prescriptive Planning Results

The fourth objective is represented in the system as a controlled planning scenario. The interface can shortlist positive-revenue products, collect user-entered demand, usable stock, protected stock, pack size, cost, supplier limits, and budget, and solve a constrained allocation. Tests cover feasibility, integer packs, stock protection, and a service-first objective. The result is useful for demonstrating how a decision can be structured, but it remains conditional on the values entered by the user.

Table 4.11 Operational inputs required for prescriptive use

Input	Current status	Why it matters
On-hand and reserved stock	Not published	Determines actual availability and stock gaps
Expiry dates and batch age	Not published	Required to measure expiry-driven wastage
Lead time and service level	Not published	Required for ROP and safety-stock calculations
Ordering and holding cost	Not published	Required for defensible EOQ
Pack size MOQ and supplier cap	User-entered assumption	Constrains feasible orders
Budget and approved demand	User-entered assumption	Determines the allocation objective and ceiling

Source: MedShield business definitions, analytics guidance, and planning interface.

No actual wastage rate can be calculated because no expiry or write-off outcome dataset is available. The proposal's target of expiry-driven wastage at or below 5 percent remains a goal

for a future operational evaluation. A claim of achievement would confuse a design objective with a measured result.

4.3 Dashboard

Section 4.3 explains the dashboard as the visible part of the decision-support system. A reviewer can follow a summary back to its period, source records, evidence label, and model or scenario status. The 2026 display follows the same rule: observed values require published transactions, while elapsed months awaiting actual records may show separately labeled estimates based on earlier years.

4.3.1 Dashboard Structure and Navigation

The MedShield dashboard is the presentation layer of the decision-support system. An authenticated user enters through Overview and can open Sales Diagnostics, Product Prioritization, Area Prioritization, Forecast Modeling, and Prescriptive Planning. Supporting views show transaction-level sales evidence and weather-source validation. A Data Upload page exists, but client CSV ingestion and publication are still planned. Navigation places the historical result, source record, assumption, and limitation within one review path.

A shared top bar carries the selected period and comparison context. Single-year and year-to-year controls draw available years from the dataset. Changing the selection refreshes related cards, charts, and tables for the same population. The header shows the loaded period and dataset state, while missing periods remain gaps rather than zero demand. For an elapsed 2026 month without an uploaded actual, the dashboard may calculate a same-calendar estimate from up to three earlier years, with recency weights of 0.60, 0.30, and 0.10 where observations exist. Chart and KPI labels distinguish estimates from actual sales, and an uploaded actual takes precedence. If an analytical source or required view is unavailable, the interface clears the affected result and shows an unavailable state.

4.3.2 Dashboard Modules and Decision Flow

Each principal module addresses a planning question. Sales Diagnostics shows recorded performance. The prioritization pages direct further review. Forecast Modeling compares baselines, while Prescriptive Planning examines a user-defined scenario. The workflow supports human review rather than automatic ordering. Figures 4.7 to 4.10 illustrate the interface; the

September development captures serve as layout evidence and do not prove current numerical outputs.

Table 4.12 Functions of the five principal dashboard modules

Module	Question answered	Main dashboard outputs	Decision use and boundary
Sales Diagnostics	How did sales, margin, quantity, and seasonality change?	KPI cards, annual and monthly trends, matched-period growth, heatmap, and STL evidence	Explains historical patterns; candidate definitions and missing periods remain visible
Product Prioritization	Which product labels contribute most historical value?	Ranked contribution, focus cohort, Pareto or ABC grouping, and product table	Focuses product review; it is not SKU approval or an automatic purchase recommendation
Area Prioritization	Which mapped areas or buyer groups require attention?	Buyer-sector, geographic, and legacy-area shares with traceable mappings	Directs commercial review; derived mappings and mixed source labels remain provisional
Forecast Modeling	How well do transparent forecasting baselines perform?	Actual-versus-holdout plots, error metrics, future baseline, and release status	Supports model comparison; there is no approved champion, interval, or external regression
Prescriptive Planning	What allocation is feasible under entered constraints?	Priority shortlist, assumption grid, feasibility result, and scenario export	Supports scenario discussion; it does not issue purchase orders or use verified inventory and expiry data

Source: MedShield Current State, Dashboard Design Logic, Analytics, and System Documentation.

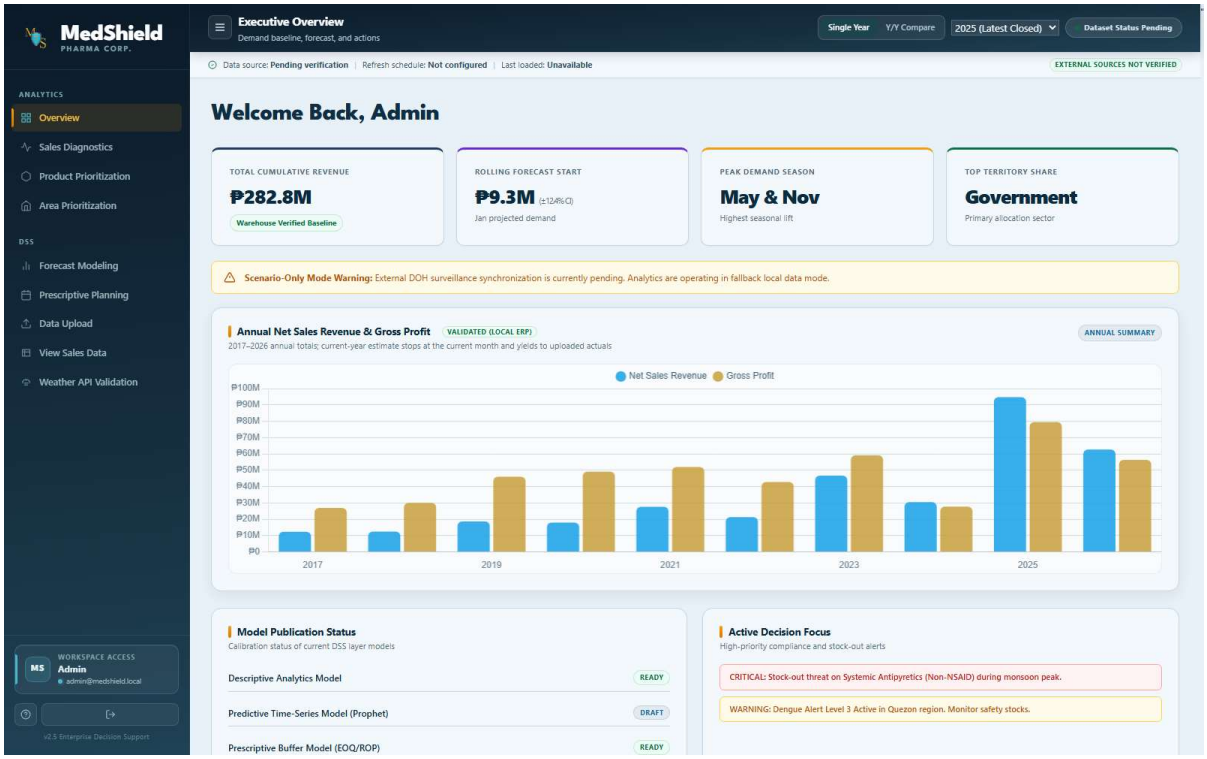


Figure 4.7 Overview dashboard development capture

Source: MedShield development interface capture dated 15 September 2026. Layout evidence only; current numerical results require a current version-bound receipt.

Overview places the selected period, sales summary, and source status at the beginning of the review. The historical source ends in December 2025. A mixed window extending into 2026 must show observed sales and gap estimates separately; a combined display is not wholly historical sales.

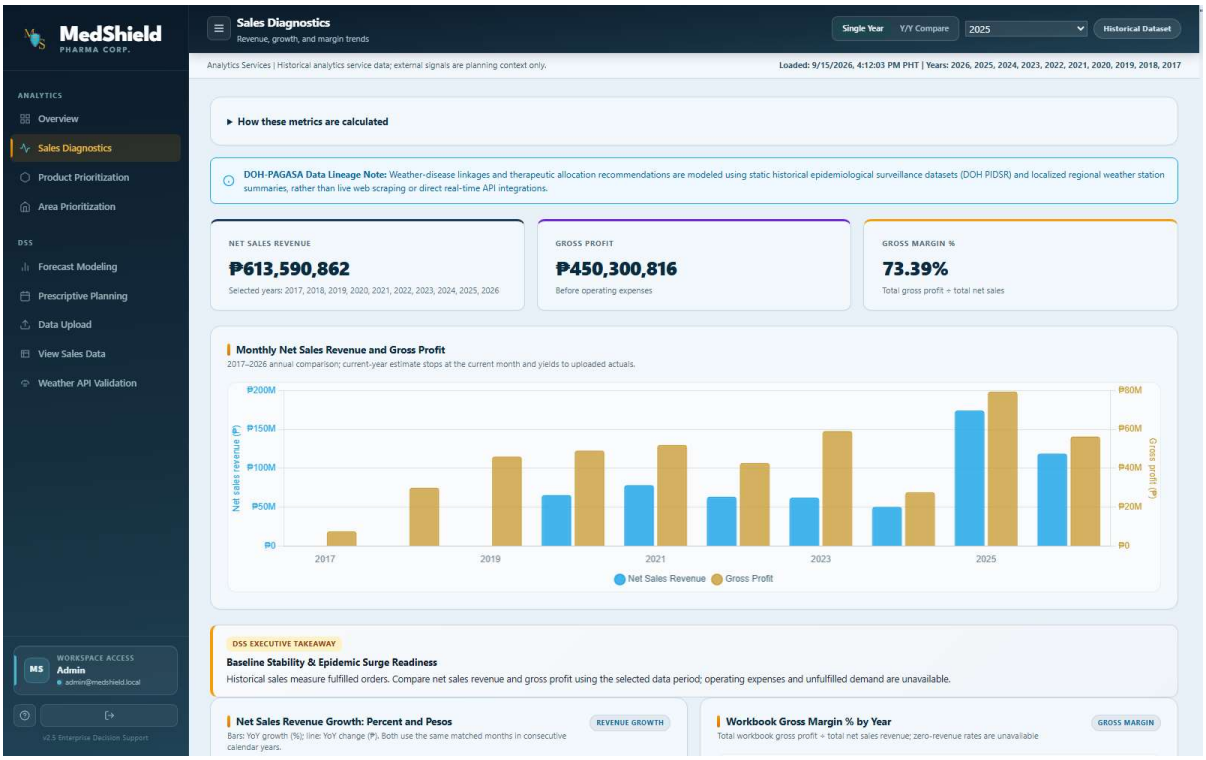


Figure 4.8 Sales diagnostics development capture

Source: MedShield development interface capture dated 15 September 2026. Layout evidence only; figures and labels require current runtime verification.

Sales Diagnostics brings year comparisons and monthly patterns into the same view. Missing months remain gaps rather than zero sales. A reviewer can use the view to locate unusual periods and then inspect the underlying transaction record.

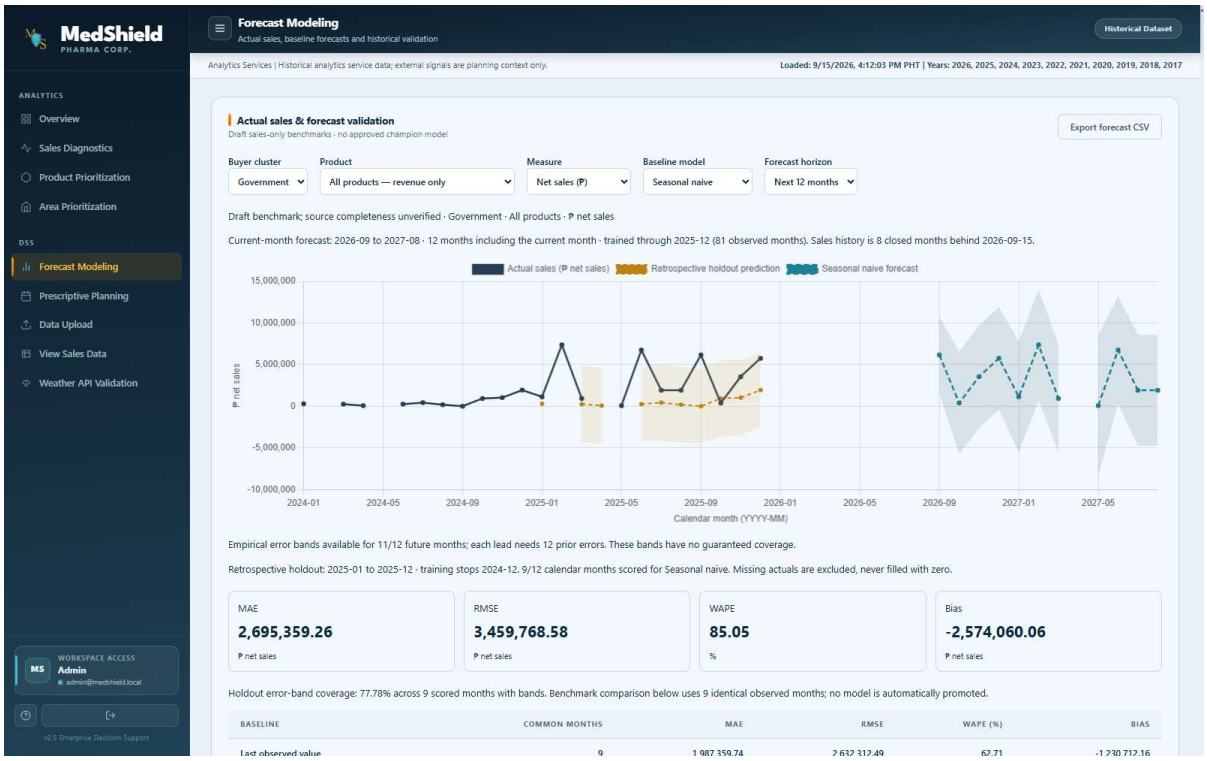


Figure 4.9 Forecast modeling development capture

Source: MedShield development interface capture dated 15 September 2026. Historical screen layout only. Any shaded forecast band in the capture is superseded by the current unsupported-interval status; no model is approved for release.

Forecast Modeling places the selected buyer, product, measure, baseline, and horizon above a comparison chart. Actual sales and retrospective holdout predictions appear apart from a future baseline. MAE, RMSE, WAPE, and bias help a reviewer judge the comparison. The older capture shows a band, but the current analytical record does not support an empirical uncertainty interval. No band or future value should be cited as an approved operational forecast.

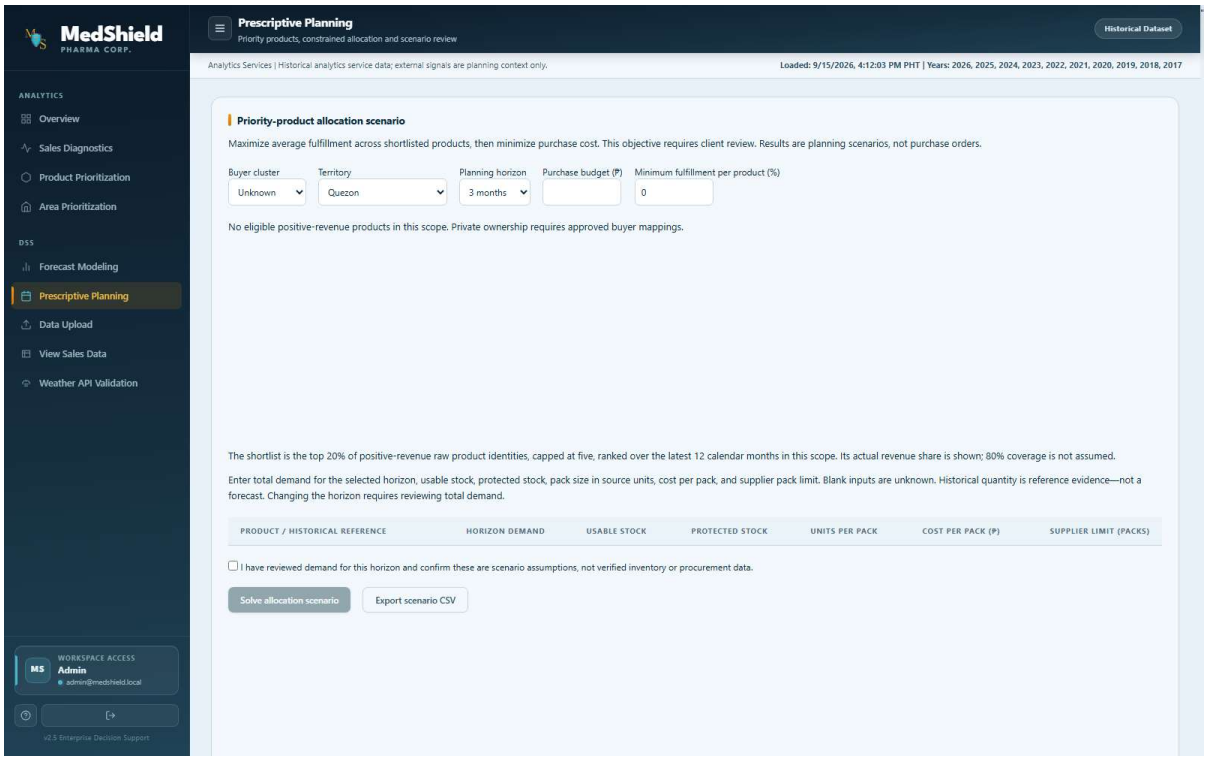


Figure 4.10 Prescriptive planning development capture

Source: MedShield development interface capture dated 15 September 2026. The displayed scope has no eligible positive-revenue products and does not show a solved allocation. Inputs remain scenario assumptions, not purchase orders.

Prescriptive Planning begins with scope and budget controls and requests demand, usable stock, protected stock, pack size, cost, and supplier limits. A confirmation control identifies entered values as assumptions rather than verified inventory records. The displayed empty product scope cannot establish solver performance; separate feasibility tests support the scenario capability.

View Sales Data supplies transaction details and provenance for inspecting a displayed total. Weather API Validation identifies official observations separately from NASA POWER proxy data. The Data Upload screen is present, but the current upload endpoint rejects submissions with a warehouse-ingestion-required response. File validation, durable ingestion status, and client-controlled publication therefore remain future work. The supporting views help users check provenance, while the upload limitation prevents a claim of completed 2026 actual-data turnover.

4.3.3 Evidence Labels and User Controls

Evidence labels help prevent overinterpretation. Actual identifies observed sales; candidate identifies data or mappings awaiting approval; estimated identifies a calculated gap

value; proxy identifies a substitute external observation; baseline identifies a forecasting comparator; blocked identifies a result unavailable under current data gates; and scenario identifies an output based on user assumptions. Loading, empty, infeasible, and unavailable states appear explicitly. The browser does not train a model, call an external weather provider directly, or write a purchase transaction. The boundaries preserve the dashboard's role as decision support for an authorized reviewer.

4.3.4 Dashboard Verification

The current QA review reports 70 of 70 publication checks passing for the version-16 historical candidate. The report establishes technical reconciliation at the review date, not client approval of product identities, area mappings, financial definitions, or a model release. The local checkout contains the earlier version-15 calculation receipt used for the detailed Chapter 4 tables; the later version-16 calculation receipt cited by QA was unavailable locally. The QA review also reports 21 backend unit tests, eight descriptive browser cases, five area and buyer-scope regressions, and six matched-period cases passing, with four retired legacy cases skipped. The sets overlap and are not combined into one test total.

Table 4.13 Validation and acceptance evidence

Validation area	Observed result	Interpretation
Publication readiness	70 pass 0 fail 0 unverified	Technical candidate checks passed; business approval still open
Descriptive population	37,728 facts 108 calendar months 98 observed months	Counts and periods reconcile; ten months remain unavailable
STL	72 reconciled months and four components	Qualifying 2020 to 2025 segment only
Backend unit tests	21 passed	Covers current backend logic including filter validation
Final descriptive browser run	8 passed	Live version-bound descriptive behaviors and exports
Area and buyer regression	5 passed	Scoped classification behavior
Matched-period and growth suite	6 passed 4 legacy skipped	Skipped tests are retired and are not counted as passes
Predictive warehouse candidate	1,092 rows and 15 evaluations persisted	Candidate experiment evidence; active model pointer remains empty

Source: MedShield System QA Review, Current State, Database, and Predictive Operations records dated 27 September 2026.

4.4 Summary of Findings

Section 4.4 answers the proposal's practical questions using the observed 2017 to 2025 record, tested system behavior, and documented limits. The 2026 gap estimates provide interim

planning context while actual transactions are pending. They do not change the historical totals or establish a measured 2026 outcome.

4.4.1 Answers to the Practical Questions

For the question of which products performed best, SPEEDA 2.5IU/0.5ML ranks first among source product labels in the current candidate. The ranking remains provisional because the approved product master is empty and aliases may divide a commercial product. Revenue-based ABC grouping directs review but does not establish clinical importance or future demand.

For next-season sales, MedShield reports retrospective baseline comparisons but cannot provide an approved operational forecast. The selected last-value method retained large errors, external disease and weather joins were not forecast-eligible, and 2026 actual sales were unavailable for a genuine 2026 holdout. The dashboard's weighted 2026 gap values are estimates for elapsed periods, not validated forecasts.

For excess or insufficient stock, sales alone cannot establish a stock position. On-hand units, reserved stock, lead times, and expiry dates were not supplied. For possible action, the planning module can test a user-entered allocation scenario and report feasibility. No scenario proves a purchase quantity, avoided expiry, or the proposal's five-percent wastage target without approved operational data.

4.4.2 Historical Patterns

The technically reconciled candidate shows ₱626.95 million in observed net sales across 2017 to 2025. Recorded activity is uneven by year and by month, with ten missing months in the early history and a large 2025 value requiring careful provenance review. May and June have the highest observed calendar-month averages. STL provides a separate 2020 to 2025 trend, seasonal, and residual decomposition without imputing the earlier gaps.

4.4.3 Forecast Comparison

On the comparable fixed-origin Government holdout, last observed value performed better than seasonal naive, but its WAPE remained 62.71 percent. In the larger offline experiment, last value also remained selected, with 83.20 percent development WAPE and 37.76 percent retrospective WAPE. Large errors, unsupported uncertainty intervals, and unapproved external joins justify keeping the model unpublished.

4.4.4 Product and Area Priorities

SPEEDA 2.5IU/0.5ML is the highest-revenue source product label in the current candidate, while GOVERNMENT is the highest-revenue derived area label. Both findings are useful for review, but neither should be treated as a final canonical SKU or a geographic market-share statement. The approved product master is empty, and the legacy area field mixes different commercial meanings.

4.4.5 Planning Scenarios

The system can produce constrained allocation scenarios after a user supplies operational assumptions. A result identifies feasible allocations under the entered values and exposes infeasible constraints. Real EOQ, ROP, safety stock, and expiry reduction require published inventory and procurement data.

4.4.6 Decision Support Value

MedShield places source coverage, KPI definitions, historical patterns, priority candidates, forecast errors, and planning assumptions in one review path. The arrangement supports demand review, but no before-and-after measure shows a quantified improvement in visibility. No REC-approved participant evaluation appears in the available record. System tests establish selected behaviors, not perceived usefulness or user acceptance.

4.4.7 Discussion of the North Star

The evidence supports the demand-visibility aim more directly than the expiry-loss aim. MedShield consolidates historical sales, preserves gaps, shows KPI bases, and displays the status of models and mappings. A reviewer can move from a historical priority to an allocation scenario. The workflow makes assumptions easier to inspect, although no before-and-after study measures a visibility gain.

The study does not demonstrate reduced expiry-driven losses. Measurement requires a baseline of expiry or write-off losses, comparable post-implementation observations, and operational inventory inputs. The current system can prepare an evaluation by recording scenario assumptions and decisions. Until outcome data exist, MedShield provides information and controls for planning without proving the financial effect of a plan.

Simple models and explicit release gates proved useful in the forecast review. Prophet configurations did not earn promotion merely through greater complexity. Last value performed

best among the tested candidates, but large errors and weak uncertainty evidence kept the model unpublished. An authorized user therefore retains the final planning decision, with the basis of any scenario available for inspection.

4.4.8 Limitations

- The Gold sales objects remain candidate publications pending named business and finance approval.
- The product master has no approved entries, so product analysis uses source labels and cannot standardize aliases or pack conversions.
- The legacy area field combines geographic, customer-type, and business-line labels; final mapping approval is still required.
- Ten months have no observed transactions. The study preserves the gaps but cannot independently determine whether missing files or absent business activity caused them.
- DOH and PAGASA rows are not approved for territory-level forecasting, while NASA POWER remains a separately labeled proxy.
- No predictive model or uncertainty interval is approved for operational publication.
- Inventory, expiry, supplier, lead-time, service-level, ordering-cost, holding-cost, and stock-position data are unavailable.
- The study contains technical and browser validation but no completed REC-approved participant evaluation or final client acceptance record.

4.4.9 Overall Synthesis

MedShield has a working historical decision-support foundation. The system reconciles 37,728 candidate sales facts, presents 2017 to 2025 performance, preserves ten missing months, decomposes a qualifying 72-month series through STL, ranks product and area labels, compares common-holdout baselines, and exposes constrained planning scenarios through an authenticated dashboard architecture. The descriptive and presentation objectives have the strongest evidence.

Important limits remain. Product identity and area mappings still need owner approval. External signals are not forecast-eligible. The selected forecast remains unpublished because its errors are high and uncertainty support is incomplete. Prescriptive outputs remain scenarios, and the expiry-wastage target has not been measured. MedShield can therefore be defended as a

historical DSS for traceable review, not as proof of an operational forecast or reduced expiry losses.

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